MALWANDLA HLONGWANE

Junior Software Developer | Full-Stack Web Developer

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PROFESSIONAL SUMMARY

Final-year Diploma in ICT (Applications Development) student at the University of Mpumalanga who builds and ships full-stack web applications. Projects include a live Node.js and MySQL marketplace with 134 automated tests, an enterprise Jakarta EE platform with CI/CD and Docker, and a security-hardened PHP MVC booking site. Skilled in REST API design, authentication and web security, relational database design, Git and automated testing. Seeking a Junior Software Developer, Graduate Developer or Software Development Internship role.

TECHNICAL SKILLS

- **Languages:** Java, JavaScript, PHP, SQL, HTML5, CSS3, Dart
- **Backend & APIs:** Node.js, Express.js, Jakarta EE (Faces, REST, EJB, JPA, JMS), Spring Boot, REST API design, MVC and layered architecture
- **Databases:** MySQL, PostgreSQL, SQLite; schema design, normalisation, indexing, transactions
- **Security:** JWT, two-factor authentication (TOTP), bcrypt/PBKDF2 hashing, AES encryption, CSRF protection, Content-Security-Policy, rate limiting, input validation
- **Testing & DevOps:** Automated testing (Node.js test runner, JUnit 5), Postman, Git, GitHub, GitHub Actions (CI), Docker, Maven, Render
- **Practices:** Agile/Scrum, SDLC, requirements (SRS), UML modelling, object-oriented design, design patterns
- **Professional:** Problem solving, teamwork on group projects, technical documentation, attention to detail, fast learner

PROJECT EXPERIENCE

Taskify | Campus Task & Equipment Marketplace

Full-stack web application | Deployed live
Tech: Node.js, Express, MySQL, JavaScript, HTML, CSS, REST API, JWT, Render

- Built and deployed a full-stack marketplace where students post tasks, hire services, buy items and rent equipment.
- Designed a 26-table MySQL schema (39 foreign keys, 76 indexes) behind a REST API with separate routes, controllers and services.
- Secured the platform with JWT authentication, TOTP two-factor sign-in for admins, bcrypt password hashing, Helmet security headers, rate limiting and server-side input validation.
- Stopped double bookings and duplicate payments under concurrent use with row-level locks and conditional updates.
- Wrote 134 automated tests across 13 test suites (Node.js test runner) covering rentals, sales, disputes and tasks, all passing.
- Integrated Cloudinary image uploads, Nodemailer email and web push notifications.

GitHub: github.com/Malwandla21-ict/taskify | **Live demo:** taskify-frontend-ol07.onrender.com

StokVault | Stokvel Contribution & Payout Platform

Enterprise Java | Group project (DICT312) for UMP Centre for Entrepreneurship, Cooperatives & Innovation
Tech: Java 21, Jakarta EE 11 (Faces, REST, EJB, CDI, JPA, JMS, Batch), Payara Server, PostgreSQL, Maven, Docker, GitHub Actions, JUnit 5

- Delivered a layered Jakarta EE application that replaces WhatsApp and notebook record-keeping for stokvels, with role-based access for treasurers, committee members and members.
- Automated payout eligibility with a scheduled Jakarta Batch job and required a second approver (four-eyes check) for high-value payouts.
- Built a SHA-256 hash-chained, tamper-evident audit log, with a PostgreSQL trigger that blocks updates and deletes.
- Protected sensitive data with PBKDF2 password hashing, AES-256-GCM encryption of ID numbers and account lockout after failed logins.

- Set up GitHub Actions CI to build the app, run 49 JUnit 5 unit tests and 48 end-to-end API checks, and build a Docker image.

GitHub: github.com/Malwandla21-ict/StokVault

UMP Events | Campus Event Booking Website

PHP web application | Started as a 2nd-year group project, since rebuilt

Tech: PHP 8.2, MySQL, PDO, MVC, HTML, CSS, JavaScript, Apache (XAMPP)

- Rebuilt a procedural PHP site into a framework-free MVC application with a router, middleware, controllers, services and repositories.
- Students browse and book events and get a digital ticket with a booking reference; admins manage events, track fill rates and export attendee lists to CSV.
- Prevented overbooking with database transactions, row locks and unique constraints.
- Hardened security against common OWASP risks: prepared statements, CSRF tokens, a strict Content-Security-Policy, secure sessions, login throttling and safe file uploads.

GitHub: github.com/Malwandla21-ict/Ump-Event-Booking-web

ADDITIONAL PROJECTS

FUEL SA | Fuel Ordering & Station Management System

Java desktop application

Tech: Java 17, JavaFX 21, FXML, CSS, SQLite (JDBC), Maven, BulkSMS REST API

- Built a desktop app where customers register vehicles and request fuel with a live cost estimate, and admins approve or reject requests from a dashboard.
- Integrated the BulkSMS REST API to send request status updates by SMS.
- Stored passwords as salted PBKDF2 hashes and kept API credentials out of source code using environment variables.

GitHub: github.com/Malwandla21-ict/FuelSA

STAMS | Student Accommodation Management System

Mobile app + REST API | In progress

Tech: Java 21, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, PostgreSQL, Flutter, Dart, Postman

- Building a Flutter mobile app and Spring Boot REST API for landlords to manage properties, rooms and student residents.
- Implementing JWT authentication with Spring Security and testing API endpoints in Postman.

GitHub: github.com/Malwandla21-ict/STAMS-Student-Accomadation-Management-System-

EDUCATION

Diploma in Information and Communication Technology (Applications Development)

University of Mpumalanga | Expected completion: 2026 | Academic average: 75–78%

Relevant coursework:

- **Web Development:** PHP (OOP, forms, sessions, authentication), MySQL with PDO, JavaScript (DOM)
- **Application Development (Jakarta EE):** Servlets, JSF, RESTful web services, JSON binding, JPA/ORM, microservices
- **Software Engineering:** SDLC, requirements engineering, UML, architectural design, Agile/Scrum, software testing